

Zinc-mediated addition of diethyl bromomalonate to alkynes for the cascade reaction towards polysubstituted pyranones and tetracarbonyl derivatives†

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Anne Miersch, Klaus Harms and Gerhard Hilt*

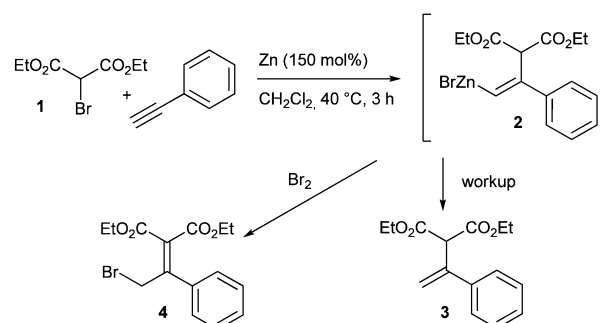
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The zinc-mediated regioselective addition reactions of diethyl bromomalonate and aromatic and aliphatic alkynes were investigated for the synthesis of vinyl malonates. When the vinyl organo-zinc intermediates were reacted with acid chlorides 2*H*-pyran-2-ones were obtained while the application of oxalyl chloride and an amine led to tetracarbonyl derivatives in a one-pot multi-step reaction sequence.

Organo-zinc species have been investigated intensively by many groups for their application in various carbon–carbon bond formation processes.¹ The zinc-mediated addition of organo-zinc species generated *in situ* to carbon–carbon double and triple bond systems has been investigated in the past as well.² The mild reaction conditions and the comparable low reactivities of organo-zinc species make them highly interesting regarding the high compatibility towards a variety of functional groups.³

Recently, we described the regiodiverse addition of benzylic bromides to alkynes mediated by varying amounts of zinc powder for the control of the double bond geometry.⁴ When the benzylic bromide was replaced by diethyl bromomalonate (**1**) the addition of the *in situ* generated organo-zinc species to phenyl acetylene led to the vinyl zinc intermediate **2** which was then quenched by electrophiles. The work-up of intermediate **2** after a reaction time of 3 h led to the mild generation of a vinyl malonate ester **3** without isomerisation of the double bond (Scheme 1). Also, the addition of the organo-zinc species was highly regioselective and only the 1,1-disubstituted alkenes of type **3** were obtained (see Fig. 1).

However, when bromine was used for the work-up of **2** the double bond migrated from a vinyl bromide to the allyl bromide position as in **4**. Interestingly, products of type **4** would be accessible by the well-known Knoevenagel condensation,⁵ whereas the thermodynamically less stable products of type **3** are not accessible under the original conditions of the Knoevenagel condensation.⁶



Scheme 1 Zinc-mediated addition of diethyl bromomalonate to phenylacetylene and quenching experiments of the intermediate.

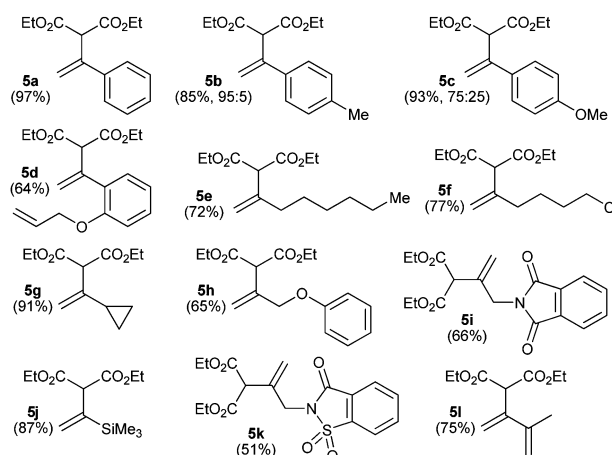


Fig. 1 Results for the zinc-mediated addition of diethyl bromomalonate **1** to alkynes.

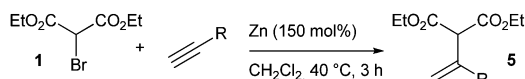
The reaction worked best in dichloromethane as solvent giving the desired product **3** in 97% yield when the alkyne was applied in slight excess (1.5 equiv.) (Scheme 2). Only 62% product was isolated when toluene was used as solvent and in this case three isomers with the same molecular mass were observed in the inseparable mixture of products. When many zinc-mediated reactions were performed in

Fachbereich Chemie, Hans-Meerwein-Str., 35043 Marburg, Germany.

E-mail: Hilt@chemie.uni-marburg.de; Fax: +49 6421 2825677;

Tel: +49 6421 2825601

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Scheme 2 Regioselective synthesis of vinyl malonates of type 5.

ethereal solvents such as diethyl ether or tetrahydrofuran,⁷ no conversion was observed in these solvents and only traces of the desired product 3 could be detected by GCMS when acetonitrile was applied as solvent. The reaction also failed when catalytic amounts of zinc powder (10 mol%) were used. However, with an excess of zinc (150 mol%) and a 1 : 1 ratio of the starting materials already 71% yield was obtained and an excess of alkyne (1.5 equiv.) was found to be optimal.

Under these reaction conditions we investigated the scope of the reaction toward a number of functionalized terminal alkynes. The results of these reactions are summarized in Fig. 1. For electron-rich phenyl acetylene derivatives (5a/5c) good yields were obtained. However, the more electron rich the arene substituent, the higher the ratio of double bond migration toward the Knoevenagel-type product (comparable to 4, Scheme 1). In the case of the 4-methoxyphenyl substituent (5c), already 25% of the double bond migrated within 24 h after isolation.⁸ Nevertheless, when aliphatic substituted alkynes were applied (5e–5g) these double bond migration products were not observed within a reaction time of 3–4 h. The yields for most products were good leaving the functional groups involved in the starting material intact.

Incorporation of heteroatoms into allylic positions in the products could be realized (5h/5i/5k) and the vinyl cyclopropane derivative (5g) and the vinyl silane derivative (5j) are also of similar interest for follow-up reactions.

In the next step of the investigation, the intermediates of type 2 were reacted with acid chlorides (6) as the most reactive carbonyl derivative for direct carbon–carbon bond formation with vinylic organo-zinc species (Scheme 3). The conversion with the acid chloride led to intermediates 7 which could not be isolated but led to a condensation reaction yielding the 2*H*-pyran-2-ones of type 8 as products.⁹

In this series of experiments the alkyne was kept constant (phenylacetylene) and the acid chloride derivative was altered to investigate the scope of the substituents (R) compatible with the reaction sequence. The results of the zinc-mediated synthesis of 2*H*-pyran-2-ones are summarized in Table 1.

The one-pot multi-step synthesis of 2*H*-pyran-2-one derivatives could be realized for a series of acid chlorides in acceptable to

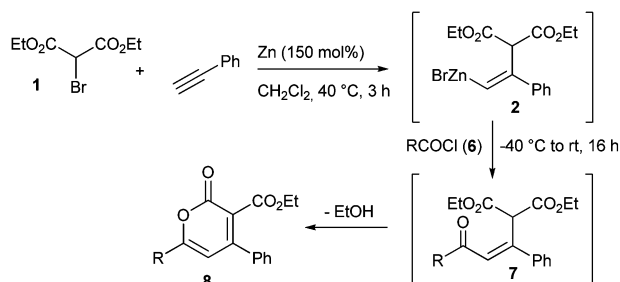
Table 1 Results for the zinc-mediated cascade reaction sequence for the synthesis of 2*H*-pyran-2-ones of type 8

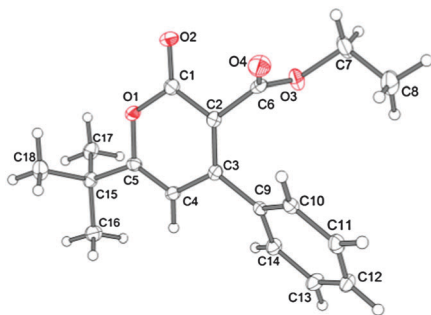
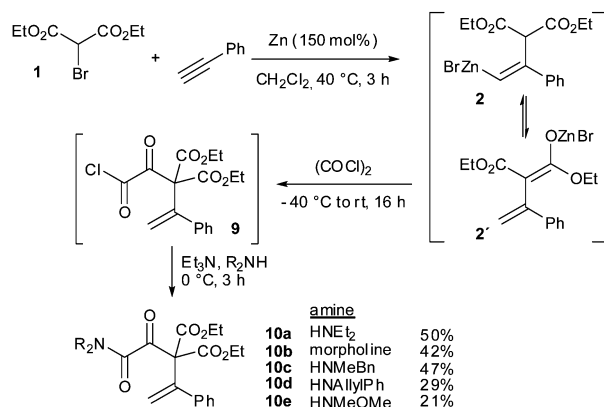
Entry	RCOCl	Product	Yield ^a (%)
1	R = Bn		76
2	R = cyclohexyl		69
3	R = <i>tert</i> Bu		66
4	R = Me		62
5	R = Et		60
6	R = 4-ClC ₃ H ₆		55
7	R = C ₁₀ H ₁₅		52
8	R = C ₃ H ₅		45
9	R = (CH ₂) ₂ CH=CH ₂		42

^a Conditions: diethyl bromomalonate (1.0 eq.), phenylacetylene (1.5 eq.), zinc dust (150 mol%), CH₂Cl₂ (1 mL), 40 °C, 3 h, then acid chloride (3.0 eq.), –40 °C–rt, 16 h.

good yields. However, the reaction seems to be limited to aliphatic acid chlorides because aromatic acid chlorides gave the desired products only in low yields.

Nevertheless, simple aliphatic acid chlorides (8d/8e), sterically hindered acid chlorides (8c/8g), as well as functionalized aliphatic acid chlorides (8f/8h/8i) are well accepted. Thereby, the construction of trisubstituted and polyfunctionalised 2*H*-pyran-2-ones in a straightforward fashion was accomplished.

Scheme 3 Cascade reaction sequence for the synthesis of 2*H*-pyran-2-ones.

Fig. 2 X-ray structure of **8c**.Scheme 4 Zinc-initiated cascade reaction sequence for the synthesis of tetracarbonyl derivatives of type **10**.

In all these cases the carbon–carbon bond formation between the acid chloride and the vinyl-zinc intermediate took place at the carbon bound to the zinc atom.

The identity of the 2H-pyran-2-ones was established by NMR techniques as well as by X-ray analysis of the crystalline products **8b**, **8c** and **8e**.¹⁰ The X-ray structure of **8c** is given in Fig. 2.

In contrast to these results, when oxalyl chloride was applied a different type of product was obtained (Scheme 4). The zinc-mediated reaction of **1** with the alkyne led to intermediate **2** which is in equilibrium with the zinc-enolate form **2'** after intermolecular migration of zinc bromide. Upon performing several deuterium labelling experiments this assumption could be verified. Detailed results can be found in the ESI.† Regular acid chlorides (**6**) attack at the vinyl-zinc position to form intermediates **7** while the more reactive oxalyl chloride reacts at the malonate position to form intermediate **9**. Finally, after addition of a secondary amine the tetracarbonyl reaction product **10** is formed.

The reaction could be realized so far for secondary amines in moderate yields. The products of type **10** are densely functionalized and initialize unexplored possibilities for follow-up transformations.

In conclusion, we were able to demonstrate that the addition of bromomalonate to triple bonds can be facilitated by zinc leading to the regioselective addition to the corresponding vinyl-malonates of type **5** in good to excellent yields. Also, the vinyl-zinc intermediate could be utilized for the synthesis of pyranones of type **8** when acid chlorides were used. In these products several additional functional groups were tolerated and electron-deficient pyranones were

accessible in a one-pot procedure. Additionally, the reaction with oxalyl chloride led to a diverse reaction sequence to form highly functionalized tetracarbonyl derivatives (**10**) with unexplored synthetic potential.

Notes and references

- (a) A. Alexakis, J. Burton, J. Vastra, C. Benhaim, X. Fournieux, A. van den Heuvel, J. M. Leveque, F. Maze and S. Rosset, *Eur. J. Org. Chem.*, 2000, 4011; (b) S. C. Berk, M. C. P. Yeh, N. Jeong and P. Knochel, *Organometallics*, 1990, 9, 3053; (c) C. M. Binder and B. Singaram, *Org. Prep. Proced. Int.*, 2011, 43, 139; (d) S. E. Denmark and J. P. Edwards, *J. Org. Chem.*, 1991, 56, 6974; (e) R. Duddu, M. Eckhardt, M. Furlong, H. P. Knoess, S. Berger and P. Knochel, *Tetrahedron*, 1994, 50, 2415; (f) C. Jubert and P. Knochel, *J. Org. Chem.*, 1992, 57, 5425; (g) S. H. Kim and R. D. Rieke, *Molecules*, 2010, 15, 8006; (h) I. Klement, M. Rottlander, C. E. Tucker, T. N. Majid, P. Knochel, P. Venegas and G. Cahiez, *Tetrahedron*, 1996, 52, 7201; (i) P. Knochel, J. J. A. Perea and P. Jones, *Tetrahedron*, 1998, 54, 8275.
- (a) K. A. Agrios and M. Srebnik, *J. Organomet. Chem.*, 1993, 444, 15; (b) G. Blay, I. Fernandez, A. Marco-Alexandre and J. R. Pedro, *J. Org. Chem.*, 2006, 71, 6674; (c) F.-Q. Li, S. Zhong, G. Lu and A. S. C. Chan, *Adv. Synth. Catal.*, 2009, 351, 1955; (d) M. Nakamura, T. Fujimoto, K. Endo and E. Nakamura, *Org. Lett.*, 2004, 6, 4837; (e) S. A. Rao and P. Knochel, *J. Am. Chem. Soc.*, 1991, 113, 5735; (f) B.-H. Tan, J. Dong and N. Yoshikai, *Angew. Chem., Int. Ed.*, 2012, 51, 9610; (g) B. M. Trost, M. J. Bartlett, A. H. Weiss, A. Jacobi von Wangelin and V. S. Chan, *Chem.-Eur. J.*, 2012, 18, 16498; (h) K. E. Schulte, G. Rücker and J. Feldkamp, *Chem. Ber.*, 1972, 105, 24; (i) M. T. Bertrand, G. Courtois and L. Miginiac, *Tetrahedron Lett.*, 1974, 1945; (j) M. T. Bertrand, G. Courtois and L. Miginiac, *Tetrahedron Lett.*, 1975, 3147.
- (a) P. Knochel, M. A. Schade, S. Bernhardt, G. Manolikakes, A. Metzger, F. M. Pillar, C. J. Rohbogner and M. Mosrin, *Beilstein J. Org. Chem.*, 2011, 7, 1261; (b) F. Langer, L. Schwink, A. Devasagayaram, P. Y. Chavant and P. Knochel, *J. Org. Chem.*, 1996, 61, 8229; (c) T. N. Majid and P. Knochel, *Tetrahedron Lett.*, 1990, 31, 4413; (d) M. Pasco, N. Gilboa, T. Mejuch and I. Marek, *Organometallics*, 2013, 32, 942; (e) M. J. Rozema, A. Sidduri and P. Knochel, *J. Org. Chem.*, 1992, 57, 1956; (f) A. Sniady, A. Durham, M. S. Morreale, A. Marcinek, S. Szafert, T. Lis, K. R. Brzezinska, T. Iwasaki, T. Ohshima, K. Mashima and R. Dembinski, *J. Org. Chem.*, 2008, 73, 5881; (g) B. M. Trost, V. S. C. Yeh, H. Ito and N. Bremeyer, *Org. Lett.*, 2002, 4, 2621; (h) X. F. Wu and H. Neumann, *Adv. Synth. Catal.*, 2012, 354, 3141; (i) L. Zhu, R. M. Wehmeyer and R. D. Rieke, *J. Org. Chem.*, 1991, 56, 1445.
- A. Miersch and G. Hilt, *Chem.-Eur. J.*, 2012, 18, 9798.
- (a) A. R. Burgoyne and R. Meijboom, *Catal. Lett.*, 2013, 143, 563; (b) G. Cardillo, S. Fabbri, L. Gentilucci, M. Gianotti and A. Tolomelli, *Synth. Commun.*, 2003, 33, 1587; (c) R. Fan, W. Wang, D. Pu and J. Wu, *J. Org. Chem.*, 2007, 72, 5905; (d) Y. Ogata and M. Tsuchida, *J. Am. Chem. Soc.*, 1959, 81, 2092.
- D. Tejedor, G. Mendez-Abt, L. Cotos and F. Garcia-Tellado, *Chem.-Eur. J.*, 2012, 18, 3468.
- (a) L. Bernardi, B. F. Bonini, E. Capito, G. Dessole, M. Fochi, M. Comes-Franchini and A. Ricci, *Synlett*, 2003, 1778; (b) D. Chen and M.-H. Xu, *Chem. Commun.*, 2013, 49, 1327; (c) D. R. Fandrick, J. T. Reeves, J. M. Bakonyi, P. R. Nyalapatla, Z. Tan, O. Niemeier, D. Akalay, K. R. Fandrick, W. Wohlleben, S. Ollenberger, J. J. Song, X. Sun, B. Qu, N. Haddad, S. Sanyal, S. Shen, S. Ma, D. Byrne, A. Chitroda, V. Fuchs, B. A. Narayanan, N. Grinberg, H. Lee, N. Yee, M. Brenner and C. H. Senanayake, *J. Org. Chem.*, 2013, 78, 3592; (d) B. Hatano, K. Nagahashi and T. Kijima, *J. Org. Chem.*, 2008, 73, 9188; (e) W. Kong, C. Fu and S. Ma, *Org. Biomol. Chem.*, 2008, 6, 4587.
- (a) Y. Fall, F. Berthiol, H. Doucet and M. Santelli, *Synthesis*, 2007, 1683; (b) P. V. Ivchenko, I. E. Nifant'ev, Y. N. Luzikov and S. G. Mkoyan, *Synthesis*, 2007, 1038; (c) S. Krompiec, N. Kuznik, M. Krompiec, R. Penczek, J. Mrzigod and A. Torz, *J. Mol. Catal. A: Chem.*, 2006, 253, 132; (d) A. M. Linsenmeier and S. Braese, *Eur. J. Org. Chem.*, 2012, 6455.
- (a) K. Cheng, A. R. Kelly, R. A. Kohn, J. F. Dweck and P. J. Walsh, *Org. Lett.*, 2009, 11, 2703; (b) Y. Kuninobu, A. Kawata, M. Nishi, H. Takata and K. Takai, *Chem. Commun.*, 2008, 6360; (c) J. Moreau, C. Hubert, J. Batany, L. Toupet, T. Roisnel, J.-P. Hurvois and J.-L. Renaud, *J. Org. Chem.*, 2009, 74, 8963; (d) S. M. Wickel, C. A. Citron and J. S. Dickschat, *Eur. J. Org. Chem.*, 2013, 2906.
- CCDC 957488 (**8b**), 957489 (**8c**) and 957490 (**8e**).